Project Title : Student Burnout Prediction Using Time-Series + Behavioral Data

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Paper 1 : **Detecting Burnout Among Undergraduate Computing Students (Yeskuatov et al., 2025)**

This research tackles survey dependency for burnout detection through passive administrative data, such as GPA, course credits, and enrollment history, as student behavioral proxies for undergraduate computing students. Machine learning algorithms were implemented for predicting three burnout dimensions: emotional exhaustion, cynicism, and academic efficacy. Exhaustion had the highest performance among the three, with 68.4% F1-score, based on temporal decline of GPA and overload of credits. For cynicism and efficacy, however, they had lower F1 scores (~0.50-0.60), due to more nuanced student behaviors. Using 1,200 students for cross-validation showed its generalizability, where feature importance was attributed more to workload intensity rather than student demographics. This research is advantageous for large-scale and real-time monitoring through university administrative systems, rather than survey-based methods with inherent biases and low response rates. However, it is limited to its domain for computing students and lacks digital logs for real-time monitoring. Future research directions suggest incorporating Learning Management System (LMS) data for more temporal student behaviors. This research is a pioneer for non-intrusive burnout detection, validating temporal trends of administrative data for timely intervention.

Paper 2 : **Early Detection of Academic Burnout Using Student Interaction Logs (Bhat, 2025)**

This research uses time series behavioral data from online learning systems, including login frequency patterns, session duration features, quiz performance trends over multiple time points, and sentiment analysis from online student forums, to facilitate early academic burnout risk detection among 800 students over two complete semesters. Various machine learning classifiers, including Support Vector Machines, Random Forest, and Long Short-Term Memory, have been used to automatically identify academic burnout risk levels, ranging from low, moderate, to high, to inform early intervention strategies. In particular, the Random Forest classifier showed excellent performance metrics, including 87% classification accuracy and an F1-score of 0.85, especially in recognizing abnormal behavioral patterns, such as sudden peaks in user inactivity, gradual decreases in user engagement levels, and increases in submission delays, through sophisticated feature engineering techniques, including rolling statistical averages and temporal

velocity calculations. The system predicts academic burnout risks two weeks prior to their escalation, enabling timely support strategies for students. In addition, ablation tests showed that behavioral log data outperformed demographic factors by 25% in terms of predictive accuracy.

Paper 3 : College Students' Mental Health Prediction Model Based on Time Series Analysis (Wang et al.)

The CS-MHP-CSTGCN framework utilizes spatio-temporal graph convolutional networks for the prediction of mental health trajectories, focusing on the prediction of burnout-related exhaustion states, based on the assessment scores of 1,500 college students over time. Comprehensive preprocessing of the data was achieved through the incorporation of normalization as well as interpolation techniques to address the irregularities caused due to the inconsistent survey collection periods. The new architecture significantly outperformed the Long Short-Term Memory baseline models, achieving 92% accuracy, as opposed to 85%, while also outperforming the Gated Recurrent Unit models through the incorporation of temporal embeddings for the tracking of sequential patterns, as well as the detection of contagion patterns through the incorporation of graph attention. Thorough cross-validation of the model, based on hold-out datasets for separate semesters, proved the robustness of the system. Some of the limitations of the system include the periodicity of the surveys, which limits the real-time applicability of the system, as well as the generalization of the results beyond the Chinese University context.

Paper 4 : Predicting Student Cognitive Overload Using Multimodal Behavioral Time-Series (2026)

This study implements Long Short-Term Memory neural networks to forecast cognitive overload—a primary precursor to academic burnout—utilizing multimodal time-series datasets from task management applications that include deadline proximity measurements, rescheduling event frequencies, task completion latency profiles, and detailed keystroke behavioral patterns gathered from 600 undergraduate student participants. Advanced feature engineering produced specialized metrics such as entropy calculations quantifying task-switching behavioral volatility together with acceleration profiles mapping procrastination tendency trajectories, resulting in an Area Under the Receiver Operating Characteristic curve performance of 0.86 while reliably detecting early warning signals between 10 to 14 days in advance through effective sequential contextual memory retention capabilities. The developed model substantially outperformed conventional static machine learning methodologies by 15% in discriminative accuracy. SHAP-based interpretability analyses consistently identified acute workload intensification surges as the most influential predictive factors. A prototype system deployed in controlled pilot studies successfully reduced overload incident rates by 22%, demonstrating meaningful real-world impact. Limitations include selection bias toward students already utilizing

productivity applications and absence of physiological signal data integration for enhanced predictive depth.

Paper 5 : Research on Education Big Data for Student's Academic Performance Analysis based on LSTM (2024)

This LSTM-centric framework effectively analyzed vast amounts of educational big data culled from Massive Online Courses, including sequences of student login patterns, video interaction patterns, quiz submission patterns, and learning activity patterns from over 10,000 students, to effectively predict the deterioration in performance that indicates the risk of burnout. Sliding Window Methodologies helped effectively model the complex temporal relationships, especially in the patterns of diminishing interest and irregular participation. This method achieved 89% accuracy in predicting at-risk classifications, which is significantly better than the results achieved through traditional ARIMA models due to its proficiency in discerning non-linear patterns. Critical factors such as inter-session gaps and jittering in the patterns of submission have been identified as key factors that significantly impact the results. However, the process of anonymization also led to the lack of demographic clarity, which is another limitation. As the LSTM method is established as an effective means of conducting temporal big data analysis, the groundwork for the extension of this method into the prediction of holistic patterns of burnout is established.

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Paper 6 : Student Burnout: A Review on Factors Contributing to Burnout Across Different Student Populations

In the paper "Student Burnout: A Review on Factors Contributing to Burnout Across Different Student Populations," the authors discuss the findings of recent studies done on the issue of burnout in the context of academia. Burnout is the long-term outcome of exposure to academic stress, which presents itself in the form of emotional exhaustion, cynicism towards studies, and ineffectiveness in academics. According to the guidelines of the PRISMA protocol, the study analyzed 38 articles published from 2019 to 2024, obtained from Google Scholar and Semantic Scholar databases. Key findings of the study reveal that factors which contribute to burnout in students include an unfavorable school environment, excessive use of digital technology, and inadequate family/social support. Factors which contribute to the prevention of burnout include a favorable education environment and adequate family/social support. The psychological factors contributing to burnout include low emotional intelligence, inadequate coping strategies, low self-efficacy, and neurotic personality, whereas curiosity and grit show low levels of burnout in students. Academic performance, demographic factors, and lifestyle factors influence burnout in students, and low academic performance, sleep deprivation, and unhealthy lifestyle habits are associated with the development of burnout in students. The majority of the studies included in the review used Maslach Burnout Inventory as a tool for measuring burnout.

Paper 7 : Time Series Analysis for Education: Methods, Applications, and Future Directions

The study, "Time Series Analysis for Education: Methods, Applications, and Future Directions," examines time series analysis in the prediction and understanding of educational outcomes. With the increasing availability of sequential educational data, time series analysis has helped identify patterns in learning outcomes. The authors discuss the application of time series analysis in educational data, which can be categorized into academic performance, behavioral learning, socio-demographic, and psychosocial learning outcomes. Time series analysis has helped solve problems such as student performance prediction, learning behaviors, trends, etc. Time series analysis, including traditional techniques such as ARIMA, exponential smoothing, has shown

promise in educational prediction outcomes, while techniques such as support vector machines, random forests, deep learning, artificial neural networks, LSTMs, etc., are being explored to improve prediction outcomes. Multimodal data analysis and large language models need to be explored to improve time series analysis outcomes.

Paper 8 : Burnout Risk Profiles in Psychology Students: An Exploratory Study with Machine Learning

The paper "Burnout Risk Profiles in Psychology Students: An Exploratory Study with Machine Learning," seeks to explore the link between psychological and lifestyle factors and burnout among university students. The study focuses on psychology students, a particular student group that has received surprisingly little attention in burnout research despite their demonstrated vulnerability to academic pressure. The study used a cross-sectional study design with 274 psychology students from Portuguese universities. The students were measured using established measures of psychological well-being, psychological distress, problems with emotional regulation, sleep quality, physical activity, diet, and burnout. Using machine learning methods such as Random Forest, Support Vector Machine (SVM), and XGBoost, the researchers established that there are unique risk profiles for burnout among the psychology students. According to the study, psychological distress, problems with emotional regulation, sleep quality, diet, and physical activity are strongly linked with an increased risk of burnout among the students. Conversely, psychological well-being is linked with lower levels of burnout. The study showed that machine learning methods, such as XGBoost, are highly accurate in classifying students into risk groups for burnout.

Paper 9 : Predicting Burnout in College Students: A Machine Learning Approach Using Decision Tree and Psychometric Data

This 2025 Tapio study aims to predict academic burnout in college students through its machine learning approach, which utilized psychological and behavioral information. It employed a quantitative predictive method with the use of the Decision Tree classifier for 291 students from a private university in the Philippines. It utilized validated measures, namely MBI-SS, DASS-21, Brief Cope, and researcher-made sleep quality measure. Results: 47.8% of the students were classified as burnt out. Results also showed that the most important features for the model, based on feature importance, were stress, coping, and sleep duration, which accounted for 30%, 26%, and 15%, respectively. Despite the moderate performance of the model, which had 51.6% recall, 59.3% F1, and 66.9% ROC-AUC, the study emphasized the significance of stress, coping, and sleep in academic burnout, which is beneficial for the application of ML, particularly for its interpretability, in the context of education, with the possibility of attaining better results through the utilization of sophisticated ML algorithms.

Paper 10 : Academic burnout detection using behavioral data analysis

The paper seeks to examine the measurement of academic burnout among university students through the application of machine learning techniques in association with the results from the psychometric survey tools. Academic burnout is defined by emotional exhaustion, cynicism, and reduced academic efficacy. Academic burnout is associated with significant negative consequences for the mental health and academic performance of the students. The Maslach Burnout Inventory – Student Survey (MBI-SS) is used to measure burnout levels among the students. The MBI-SS measures emotional exhaustion, cynicism, and reduced professional efficacy among the students. The traditional three-factor model of the MBI-SS was confirmed through the application of the CFA model to the dataset collected from the 2,649 medical students. The issue of burnout labeling is examined through the application of the following rules: (i) a 'clinically inspired' rule, and (ii) the 'sample relative tertile' rule. The 'micro-screener' was developed through the application of the cross-validation-based feature selection method, with the aim of reducing the overall burden of the original instrument while retaining the overall power of the original instrument. The paper applies various machine learning techniques, including logistic regression, random forest, support vector machines, XGBoost, and LightGBM, for the prediction of burnout levels among the students.

Paper 11 : Modelling an Innovative Machine Learning Model for Student Stress Forecasting

Biswas, in 2024, developed an advanced machine learning approach for predicting student stress in an educational environment. According to the paper, the increasing academic demands and environmental factors have significant effects on the mental health of students. Conventional approaches for stress detection, such as surveys, fail to provide precise predictions. Therefore, the paper aims to introduce an efficient approach for stress prediction. For this purpose, the author suggests the use of the Fine-tuned Squirrel Search-driven Light Gradient Boosting Machine (FSS-LGBM) approach for the prediction of stress. The paper uses the Kaggle dataset, which consists of approximately 20 factors related to stress, including psychological, physiological, social, environmental, and academic factors. Min-max normalization is used for the features to increase the efficiency of the approach. According to the results, the suggested approach, FSS-LGBM, is more efficient than the MLP, DL, DT approaches, with 98% accuracy, 99% recall, 97% precision, and 97% F1-score.

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Paper 12 : Predicting Student Burnout and Stress Using Machine Learning Models

The paper explores the transition from reactive to proactive student wellness interventions by leveraging machine learning to detect early warning signs of academic exhaustion. This research identifies a critical gap in traditional methods, such as teacher observations and self-reports, which often fail to capture stress until a student is already in a state of crisis. By adopting a mixed-methods approach, the study integrates quantitative data including academic records, attendance logs, and Learning Management System (LMS) engagement metrics with qualitative self-report benchmarks to validate model accuracy. The framework focuses on identifying "behavioral drift," where sudden drops in participation frequency, late-night submission patterns, and inconsistent assignment performance serve as primary temporal predictors. A core contribution of this work is the development of a "risk-scoring" system that categorizes students into tiered intervention zones, allowing educators to deploy targeted support such as workload adjustments or counseling before burnout becomes debilitating. However, the study acknowledges significant implementation hurdles, particularly regarding the ethical handling of sensitive mental health data, the potential for algorithmic bias, and the necessity of maintaining "human-in-the-loop" oversight. Ultimately, this research provides a comprehensive roadmap for educational institutions to build data-driven, non-intrusive support systems that prioritize long-term student well-being over simple performance metrics.

Paper 13 : EMBRACE: Explainable Multitask Burnout Prediction Using Adaptive Deep Learning

The paper introduces a sophisticated deep learning framework designed to move beyond black-box predictions by providing clinical explainability for burnout. This research addresses the high-stress environment of medical residency by utilizing multimodal time-series data from wearable sensors specifically monitoring heart rate, sleep patterns, and physical activity to predict future burnout states. The technical core of the study is the EMBRACE framework, which utilizes an Adaptive Multitask Deep Learning approach to simultaneously handle burnout classification and secondary tasks like stress level estimation. In terms of performance, the model achieved a high level of accuracy, reaching an F1-score of 88.2% and an Area Under the ROC Curve (AUC) of 0.91, significantly outperforming traditional baseline models like Random Forest and standard LSTMs. A pivotal contribution of this research is the integration of SHAP (Shapley Additive Explanations), which provides transparency by ranking which physiological

features (e.g., REM sleep duration or heart rate variability) contributed most to a specific burnout flag. While the study is highly effective within the clinical domain of resident physicians, its reliance on continuous wearable data presents challenges for large-scale student populations due to hardware costs and data privacy concerns. Nevertheless, it establishes a high-performance benchmark for the field, demonstrating that multitask learning combined with Explainable AI (XAI) can create highly reliable and actionable early warning systems for mental health.

Paper 14 : Explainable Artificial Intelligence for Predictive Modeling of Student Stress

This research addresses the lack of affordable and interpretable tools for stress detection in higher education by developing a cost-effective, survey-based classification framework integrated with Explainable AI (XAI). The study shifts away from expensive physiological sensors, instead utilizing a supervised machine learning pipeline to classify student stress levels based on academic, social, and lifestyle factors. Six algorithms were benchmarked: Logistic Regression, SVM, Decision Tree, Random Forest, Gradient Boosting, and XGBoost with hyperparameters optimized through grid search and cross-validation. The Random Forest model achieved the highest performance with an accuracy of 90% and an F1-score of 0.89, demonstrating superior robustness in handling non-linear educational data. To overcome the "black-box" nature of high-performing models, the researchers implemented SHAP (SHapley Additive explanations) and LIME (Local Interpretable Model-agnostic Explanations). These XAI techniques identified "Teacher-Student Relationship," "Anxiety Level," and "Sleep Quality" as the most significant predictors of academic stress. While the study relies on self-reported data which may introduce response bias, the high accuracy and transparency of the model provide a scalable blueprint for university administrators to implement targeted, evidence-based mental health interventions. This work is significant for its dual focus on high predictive power and human-readable explanations, ensuring that model outputs are actionable for educators and counselors.

Paper 15 : Analyzing and Identifying Predictable Time Range for Stress Prediction

This research paper addresses the inherent unpredictability of human psychological states by integrating mathematical chaos theory with a deep learning framework. Moving beyond traditional linear models, the study utilizes phase space reconstruction to transform seemingly disordered, one-dimensional stress sequences into a high-dimensional space, revealing the underlying nonlinear dynamics and the "butterfly effect" inherent in stress systems. This approach allows the researchers to mathematically identify a "predictable time range," establishing a specific window where short-term stress behavior remains reliable for intervention. To handle the complexity of these reconstructed sequences, the authors developed a deep neural network architecture featuring a two-layer attention mechanism focusing on both dimensional and temporal features to simulate the stress system's state. The methodology was rigorously validated using the longitudinal, multi-modal Tesserae dataset, focusing on both 2-label and 3-label classification tasks.The integrated model demonstrated superior performance, outperforming "pure" deep learning models by over 2% and traditional chaos-based methods by over 8% in prediction accuracy. While the reliance on longitudinal sensing data requires consistent data streams, the study’s significance lies in its ability to quantify the limits of

predictability in mental health. By combining the structural insights of chaos theory with the predictive power of attention-based deep learning, the work provides a robust, scientifically grounded blueprint for clinicians to implement timely, evidence-based stress management strategies before psychological strain transitions into chronic illness.

Paper 16 : Students' stress prediction and explainable analysis based on improved decision trees

This research addresses the critical need for accurate and interpretable student stress monitoring by developing a machine learning framework based on an enhanced decision tree algorithm. The study shifts away from traditional diagnostic methods to a data-driven approach, comparing nine machine learning models including Logistic Regression, Support Vector Machine, and Random Forest to identify the optimal base classifier. To maximize predictive performance, the researchers introduced the Harris Hawks Optimization (HHO) algorithm to fine-tune the decision tree, resulting in an improved HHO-DT model that achieved a high prediction accuracy of 0.927. To ensure the model's outputs are actionable for educators, the study implemented the Shapley Additive Explanations (SHAP) model, providing transparency into the "black-box" predictions. This explainable AI (XAI) analysis identified blood pressure, social support, and depression as the most significant predictors of student stress levels. The high accuracy and interpretability of this framework provide a scientific basis for mental health educators and parents to implement targeted, evidence-based interventions to promote student well-being.

Paper 17 : A MENTAL HEALTH DETECTION MODEL OF COLLEGE STUDENTS UNDER THE BEHAVIOR OF LEARNING BURNOUT

This research addresses the escalating issue of academic burnout and its impact on the mental health of college students by proposing a specialized detection model based on electroencephalography (EEG) signals. Shifting from purely subjective surveys to objective physiological monitoring, the study utilizes an experimental framework where EEG data is collected from students exhibiting signs of burnout to identify specific neurophysiological patterns associated with mental health decline. The methodology centers on a hybrid deep learning architecture, specifically a Convolutional Neural Network (CNN) combined with a Stacked Long Short-Term Memory (LSTM) network and an attention mechanism, designed to capture both the complex spatial features and temporal dependencies of brain activity. Results indicate that this integrated approach significantly enhances the accuracy of mental health classification, effectively distinguishing between different levels of psychological strain induced by academic pressure. The significance of this work lies in its provision of a non-invasive, data-driven diagnostic tool that offers higher precision than traditional self-reporting methods, enabling university mental health professionals to identify at-risk students earlier and implement more effective, personalized clinical interventions.

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Paper 18 : Burnout Prediction: A Comparative Analysis of Machine Learning and Deep Learning Approaches

Mohd Zaini et al. (2025) focuses on predicting burnout by using artificial intelligence techniques and compares the performance of machine learning and deep learning models in this paper. Burnout is a psychological condition caused by prolonged stress, which often leads to emotional exhaustion, low motivation, and reduced productivity. It commonly affects individuals working or studying in demanding environments. The researchers aimed to determine which type of model is more effective at detecting burnout at an early stage. Before training the models, the data went through several preprocessing steps, including text cleaning, tokenization, and removal of irrelevant information. The study evaluated both machine learning and deep learning techniques. Machine learning models included commonly used classification algorithms, while deep learning approaches involved Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), and Long Short-Term Memory (LSTM) networks. Model performance was measured using standard evaluation metrics such as accuracy, precision, recall, and F1-score. The results showed that deep learning models generally performed better when handling complex textual data, although some machine learning models also achieved competitive results. The research suggests that AI-based prediction systems could help organizations and educational institutions identify individuals experiencing burnout earlier and provide proper mental health support.

Paper 19 : Machine Learning-Based Prediction of Mental Well-Being Using Health Behavior Data from University Students

This study by Rahman et al. (2023) explores whether machine learning can help predict the mental well-being of university students by analyzing their health and lifestyle behaviors. Mental health concerns among students have grown significantly in recent years, especially after the COVID-19 pandemic. To study this issue, the researchers used survey data collected from 17 universities in Southeast Asia. The dataset included information about students' physical activity, body mass index (BMI), academic performance (GPA), sedentary time, and demographic details.

The researchers applied several machine learning models, including Generalized Linear Models, k-Nearest Neighbors, Naïve Bayes, Neural Networks, Random Forest, Recursive Partitioning, Bagging, and Boosting methods. Each model was trained and tested to see which one could most accurately predict students who might experience poor mental well-being. The results showed that Random Forest and AdaBoost produced the best prediction performance. The study also identified the most important factors influencing mental health. These included weekly sports activity, BMI, GPA, sedentary hours, and age. The findings suggest that machine learning can successfully identify patterns related to students' mental health. Such predictive models could help universities recognize students who may be struggling and provide support earlier, potentially improving overall student well-being.

Paper 20 : Machine Learning Based Student Mental Health Predictor

Shetty et al. (2022) proposes a machine learning approach for predicting mental health risks among students. Mental health issues such as stress, anxiety, and depression are becoming increasingly common in academic environments. Many students experience these problems silently, which can negatively affect both their studies and personal lives. Because of this, early identification of students who may be struggling is very important. To build the prediction system, the researchers collected psychological, academic, and behavioral information from students across different academic fields. Unlike some previous studies that focused only on specific departments, this dataset included students from a wider range of backgrounds, which improves the reliability of the results. Several machine learning algorithms were tested, including Decision Tree, Random Forest, k-Nearest Neighbors (KNN), and Support Vector Machine (SVM). These models were trained to classify whether a student might face mental health difficulties. Among the tested models, Random Forest showed the best performance in predicting mental health risks. The results indicate that machine learning can detect patterns related to stress and emotional challenges among students. The study suggests that such systems could assist teachers, counselors, and universities in identifying vulnerable students earlier, allowing them to provide timely support and prevent more serious mental health problems.

Paper 21 : Predicting Student Performance Using Mental Health and Linguistic Attributes with Deep Learning

In this research Venkatachalam & Sivanraju (2023) examines how deep learning can be used to predict student academic performance by considering mental health and communication patterns. Many traditional prediction models focus mainly on academic records or demographic information, but they often overlook emotional and psychological factors that can influence learning outcomes. To address this gap, the researchers proposed a deep learning framework called SADDL (Student Accomplishment prediction using Distinctive Deep Learning). The

model combines multiple types of student data, including academic records, demographic information, and social media posts related to academic life. The textual data from social media were analyzed using Latent Dirichlet Allocation (LDA) to identify emotional and psychological patterns in students' language. The deep learning model uses Long Short-Term Memory (LSTM) to capture time-based patterns in data, Deep Convolutional Neural Networks (DCNN) to detect relationships between features, and a Multi-Layer Perceptron (MLP) to produce the final prediction. The experiments showed strong results. The model achieved 91.71% accuracy with an 80/20 data split and 89.3% accuracy with a 70/30 split, outperforming several traditional machine learning methods. Overall, the study highlights that mental health signals and linguistic patterns can improve the prediction of student academic success.

Paper 22 : Prediction of Mental Health Problem Using Annual Student Health Survey: Machine Learning Approach

In this study Baba & Bunji (2022) investigates how machine learning can be used to predict mental health problems among university students based on annual health survey data. Universities often collect health information from students, but it can be difficult to determine which students may need counseling or psychological support based only on survey responses. The dataset used in this research came from University A in Japan, where 3,561 undergraduate students completed health surveys in 2020 and 2021. The researchers designed two prediction experiments. In the first experiment, the model predicted mental health problems in 2020 using survey responses and demographic information from the same year. In the second experiment, the model predicted mental health outcomes in 2021 using similar data. Several machine learning models were tested, including Logistic Regression, Elastic Net, Random Forest, XGBoost, and LightGBM. Among these, LightGBM achieved the best performance, with very high predictive ability and Matthews Correlation Coefficient (MCC) values reaching up to 0.986. The study also examined response behavior data, such as response time and timestamps, but these features did not significantly improve prediction accuracy. The findings show that survey questions related to campus life, anxiety, and future concerns were strong indicators of mental health problems. The study suggests that machine learning could help universities identify students who may need mental health support earlier.

Combined Literature Review :

The issue of academic burnout among students has been a major concern for educational institutions in recent years. The pressure of academic activities, the digital classroom, and lifestyle have made students susceptible to psychological pressure, emotional exhaustion, and a reduction in their academic performance. The traditional approaches to the identification of academic burnout include the administration of surveys and the observation of students by their teachers. These approaches have helped researchers gain some insight into the issue of academic burnout. However, the issue of academic burnout among students has been a major challenge for researchers to identify using machine learning.

Some research works have emphasized the importance of identifying the factors that cause student burnout. An extensive review of the research works on student burn out revealed that student burn out was generally associated with emotional exhaustion, cynicism towards academic activities, and reduced academic efficacy. Environmental factors such as high workloads, unfavorable academic environment, and overuse of digital technologies were also major factors that contributed to student burn out. Psychological factors such as low emotional intelligence, coping mechanisms, sleep deprivation, and lack of social support were also significantly associated with student burn out.

Due to the increasing amount of educational data, researchers have begun employing machine learning methods in finding patterns related to student stress and burnout. Several predictive models have been proposed based on academic records, behavioral data, and psychological surveys. For instance, researchers have employed university data related to GPA, credit hours, and enrollment history in detecting burnouts among computing students. This demonstrates that changes in academic performance and workload intensity are possible indicators of burnouts. However, it is also possible that these models might fail in recognizing more nuanced behavioral patterns associated with students' motivation or engagement.

Other studies have also explored behavioral data obtained from online learning environments. For instance, interaction logs from Learning Management Systems (LMS), such as the frequency of logins, the length of sessions, submission patterns of assignments, and even sentiment analysis of forum posts, have been found to be useful in the prediction of burnout. Random Forest, Support Vector Machines, and Long Short-Term Memory models have been found to perform well in the prediction of abnormal behavioral patterns. These models have been able to predict the risk of burnout two weeks before the onset of severe symptoms in several experiments, allowing universities to offer support to students.

Time series analysis is also being recognized as an essential method for analyzing educational data. This is because the behavior of students is dynamic over time. LSTM networks and spatio-temporal graph convolutional networks are being implemented for analyzing patterns in academic performance and psychological tests. These models are highly efficient in dealing with

large educational datasets. Some frameworks have achieved high accuracy in predictions by analyzing learning activities, participation patterns, and emotional states over several semesters. However, most of these systems are still based on survey patterns and are unable to provide real-time services.

Besides behavioral and academic data, there have been several works using multimodal data sources. For instance, wearable devices used to track sleep patterns, heart rate variability, and physical activity levels have been used to identify stress and burnout among medical students. Deep learning architectures using multitask learning along with explainable artificial intelligence have been found to have high predictive accuracy while providing insights into the major physiological factors affecting the level of burnout. Though the models have shown promising results, there are several challenges associated with the models, especially for large-scale deployment, which requires the use of special hardware.

Explainable artificial intelligence (XAI) has emerged as an important feature for student mental health prediction models. Although machine learning models can provide accurate predictions, it has often been difficult to understand the predictions made by such models. To address this, several research works have incorporated SHAP and LIME techniques, which help understand the predictions made by the models. Using such models, it is possible for the researcher/educator to understand which factors, such as the quality of sleep, anxiety, teacher-student relationships, and intensity of workload, are impacting the predictions most. This helps increase the trust and usage of such models for prediction purposes within universities.

Another important research direction involves combining various data sets to create more robust predictive models. Some of these models combine academic records, lifestyle, social media texts, and linguistic features to improve the accuracy of predictions. Deep learning techniques, including long short-term memory networks, convolutional neural networks, and multilayer perceptrons, are also being used to analyze these varied data sets. Experimentation results show that including mental health signals and communication features can improve the accuracy of predicting academic performance and burnout.

Despite the advances achieved in the last few years, there are a few challenges in this research area. Some of the models are based on specific data sets collected from a specific institution or a specific domain of study. There are also ethical issues that are involved with regard to the handling of student data. In addition, whereas machine learning can identify risky patterns, intervention strategies require human expertise from educators, guidance, and psychological professionals.

Overall, the existing literature suggests that there is considerable potential for the early detection of student burnout and other mental health issues through the help of machine learning and data-driven approaches. By examining the academic performance, behavioral patterns, psychological factors, and physiological parameters, it has been found that it is possible for such

systems to detect at-risk students even before the occurrence of severe burnout. Future research directions should focus on the incorporation of multiple data types, the interpretability of the models, and the design of such models that are scalable enough for effective functioning in different educational settings.

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